# Propagating Vision Uncertainty into Spatial Simulation: Evaluating Classical Pedestrian Density Estimation as an Input for Agent-Based Street-Vendor Models

Muhammad Ibrahim Asghar
Department of Computer Science / Artificial Intelligence
COMSATS University Islamabad, Wah Campus

Muhammad Ahsan Khan
Department of Computer Science / Artificial Intelligence
COMSATS University Islamabad, Wah Campus

**Abstract—Street-vendor simulations require spatial representations of pedestrian demand, but errors in camera-derived inputs may alter the locations selected by simulated agents. This paper proposes a controlled framework for studying how pedestrian-localization errors propagate from a classical computer vision pipeline into an agent-based street-vendor model. Static street photographs are processed using contrast enhancement, image upscaling, and histogram-of-oriented-gradients features with a pretrained support vector machine. Detected bounding boxes are converted to estimated foot points, assigned to a 33 × 33 lattice, and smoothed to produce relative occupancy-based attractiveness maps. A standardized data interface transfers these maps to a NetLogo model in which 25 vendors balance local attractiveness against nearby competition and relocate through a boundedly rational search rule. The planned evaluation compares manually annotated reference maps, vision-derived maps, and uniform and dual-peak synthetic baselines across three competition regimes. Count error, cell-level spatial error, and hotspot overlap characterize upstream fidelity; nearest-neighbor clustering and reference-map opportunity capture characterize downstream effects. The framework separates similarity in aggregate clustering from agreement in the locations of attractive sites. It also explicitly distinguishes instantaneous occupancy from pedestrian flow and recognizes the geometric limitations of uncalibrated image-space mapping. The contribution is a reproducible study design for evaluating the decision relevance of classical vision errors, rather than a claim of improved detection accuracy. Empirical outcomes remain to be established through the proposed experiments.**

**Index Terms—Computer vision, pedestrian detection, agent-based modeling, NetLogo, spatial competition, uncertainty propagation.**

## I. INTRODUCTION

Pedestrian activity provides a useful starting point for representing the attractiveness of street-vending locations. However, attractiveness is not directly equivalent to profitability: customer preferences, product type, access restrictions, and trading costs may also influence site selection. This project therefore treats pedestrian presence as a stylized spatial opportunity signal rather than a complete economic model. Its objective is to examine whether imperfections in that signal change the spatial outcomes of a simulated market.

Computer vision and agent-based modeling offer complementary ways to investigate this problem. On the perception side, histogram-of-oriented-gradients (HOG) features and a support vector machine (SVM) provide an established approach to human detection [1]. On the simulation side, NetLogo supports autonomous agents interacting within spatially heterogeneous environments [2]. A modular connection between the two allows estimated pedestrian locations to become patch-level inputs for vendor decisions. Importantly, "training-free" here means that the project does not train a new detector; the SVM supplied with the software is already pretrained.

The scientific challenge is not simply to connect image-processing software to a simulator. Detection errors may be spatially concentrated rather than independent across the scene. Small pedestrians, partially occluded groups, and visually ambiguous regions can produce unequal localization quality. A detector may consequently preserve the broad pattern of pedestrian presence, or it may shift the strongest apparent hotspot to another part of the image. These alternatives can lead to different vendor placements even when total counting errors are similar. Downstream decision quality therefore requires assessment beyond image-level detection counts.

The proposed pipeline applies contrast-limited adaptive histogram equalization and image upscaling before multiscale HOG–SVM detection. The lower midpoint of each retained bounding box approximates a pedestrian's ground-contact point. Points are assigned to a 33 × 33 lattice, smoothed, and scaled to produce relative patch attractiveness. A NetLogo model places 25 vendors on this surface. Each vendor weighs local attractiveness against nearby competitors and searches a limited set of alternative patches after repeated low-utility outcomes. These are controlled modeling assumptions, not empirically established rules of vendor behavior.

Three research questions guide the study. RQ1: How well do vision-derived maps preserve cell-level pedestrian distributions and high-attractiveness hotspots relative to manual annotations? RQ2: How do these map differences change final vendor clustering and reference-map opportunity capture? RQ3: How does increasing the competition penalty affect sensitivity to upstream localization errors? The evaluation compares annotated and HOG-derived inputs with uniform and dual-peak synthetic surfaces under otherwise matched simulation settings.

Two directional hypotheses are proposed without assuming their confirmation. H1 predicts that spatially uneven detection failures reduce hotspot agreement with the annotated reference. H2 predicts that vendors using the vision-derived surface can exhibit clustering comparable to reference-driven vendors while capturing less annotated opportunity. Neither a fixed recall threshold nor a statistically significant clustering effect is presumed. Stronger competition may redistribute vendors away from erroneous peaks, but whether this improves opportunity capture remains an empirical question.

The contributions are threefold: a reproducible image-to-lattice interface; a controlled comparison separating perception error from simulation rules; and a joint evaluation of spatial clustering and reference-map opportunity capture. Scope is restricted to static occupancy-based inputs and stylized vendor relocation. The work does not infer pedestrian flow rates, purchasing behavior, validated market equilibria, or actual vendor revenues. Because experimental output is not provided, this paper presents the motivation and literature-grounded study design rather than completed empirical findings.

## II. LITERATURE REVIEW

### *A. Classical and Deep Pedestrian Detection*

Dalal and Triggs established HOG descriptors as an influential representation for human detection [1]. Local gradient-orientation histograms, normalized over neighboring blocks, encode shape information that can be classified using a linear SVM. Earlier work by Viola and Jones demonstrated efficient cascaded feature evaluation for face detection [3]; that contribution should not be confused with a pedestrian-specific benchmark. Together, these approaches illustrate the classical strategy of combining engineered visual features with discriminative classifiers.

Deep detectors subsequently introduced learned feature representations and different localization strategies. Faster R-CNN incorporates a region proposal network within a two-stage detection architecture [4], whereas the original YOLO formulation predicts detections through a unified single-stage network [5]. Density-regression approaches address a related but distinct task: the multi-column network of Zhang et al. accommodates variation in crowd scale [6], while CSRNet uses dilated convolutions for congested-scene density estimation [7]. These papers support distinguishing individual localization from density estimation; they do not establish that every deep model outperforms every classical detector on the proposed local scenes.

HOG–SVM is selected here as a transparent pretrained baseline, not as a claim of superior accuracy or universally lower computational cost. Its conventional $64 \times 128$ detection template motivates attention to image scale, while upscaling cannot reconstruct visual detail absent from the source. Occlusion and clothing-related silhouette variation are candidate explanations to investigate, not proven causes of errors in an untested scene. The relevant gap is therefore task-specific: detector accuracy alone does not establish whether its spatial output is adequate for downstream vendor decisions.

### *B. Spatial Representation and Perspective*

Crowd-density methods demonstrate that pedestrian activity can be represented as a spatial field rather than only as a list of boxes [6], [7]. PETS2009 also provides a benchmark context for evaluating surveillance tasks involving pedestrians [8]. For the present study, a discrete lattice and Gaussian smoothing offer a deliberately simple representation: each annotated or detected person contributes to a local occupancy surface. This representation differs from learning a density regressor, and its smoothing bandwidth is a modeling choice that requires sensitivity checks.

Assigning a box's lower midpoint instead of its center reduces the obvious vertical offset between torso location and approximate ground contact. Nevertheless, this operation is an image-space approximation, not a calibrated ground-plane projection. Perspective geometry means equal image cells need not represent equal physical areas [9]. Accordingly, a hotspot on the unrectified lattice must be interpreted as an image-space occupancy hotspot. Ground-plane homography would require appropriate correspondences and a defensible planar-scene assumption.

Normalization introduces a second distinction. Dividing a smoothed map by its maximum preserves relative shape but removes absolute magnitude; it is maximum-based scaling, not general min–max normalization unless the minimum is zero. Raw counts and relative maps should therefore be retained separately. A top-20% hotspot measure also requires explicit handling of ties and empty maps. The literature motivates spatial representations, but the unresolved issue here is whether such representation choices preserve enough information for consistent downstream decisions.

### *C. Agent-Based Spatial Competition*

Hotelling's spatial competition model offers a foundational account of location incentives in a simplified market [10]. However, the proposed vendor utility is not a direct implementation of Hotelling's pricing and consumer-choice assumptions. It is a stylized local interaction rule combining patch attractiveness and a competition penalty. Geographical agent-based modeling provides the broader methodological setting for heterogeneous spatial environments [11], while Wilensky and Rand explain the construction and analysis of agent-based systems using NetLogo [2]. These foundations

motivate the approach without validating any particular vendor rule.

Clear reporting is especially important when environmental inputs are compared across stochastic simulations. The updated Overview, Design concepts, and Details protocol provides a structured basis for specifying entities, scheduling, initialization, and submodels [12]. In this project, the same behavioral rules, competition settings, run lengths, and replication strategy should be applied across map conditions. A finite 200-tick outcome should be described as a final configuration unless convergence is independently demonstrated; stochastic relocation does not automatically establish equilibrium.

Clark and Evans introduced nearest-neighbor distance as a measure of spatial relationships [13]. Its index provides a useful descriptive indicator of clustering or dispersion relative to a theoretical random pattern. However, a finite, non-wrapping lattice with one vendor per occupied patch differs from an unbounded continuous random process. Statistical conclusions therefore require edge-aware analysis or simulation-based null distributions matched to the model domain, rather than treating an index below one as sufficient evidence of significance.

Clustering also does not identify whether vendors occupy desirable locations. The proposed opportunity-capture ratio evaluates final placements on the same annotated reference surface, comparing vision-driven placements with reference-driven placements. It is a relative occupancy-based score, not measured revenue or a guarantee of optimality. The denominator may be zero, and the ratio may exceed one because the reference-driven simulation itself uses bounded search. Taken together, the literature supports a focused research opportunity: jointly evaluate map fidelity, emergent spatial structure, and reference-based location quality, without claiming that the broader vision-to-simulation field lacks prior work.

## REFERENCES


[1] N. Dalal and B. Triggs, “Histograms of oriented gradients for human detection,” in Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR), vol. 1, 2005, pp. 886–893, doi: 10.1109/CVPR.2005.177.

[2] U. Wilensky and W. Rand, An Introduction to Agent-Based Modeling: Modeling Natural, Social, and Engineered Complex Systems with NetLogo. Cambridge, MA, USA: MIT Press, 2015.

[3] P. Viola and M. Jones, “Rapid object detection using a boosted cascade of simple features,” in Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR), vol. 1, 2001, pp. I-511–I-518, doi: 10.1109/CVPR.2001.990517.

[4] S. Ren, K. He, R. Girshick, and J. Sun, “Faster R-CNN: Towards real-time object detection with region proposal networks,” in Advances in Neural Information Processing Systems, vol. 28, 2015, pp. 91–99.

[5] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, “You only look once: Unified, real-time object detection,” in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2016, pp. 779–788, doi: 10.1109/CVPR.2016.91.

[6] Y. Zhang, D. Zhou, S. Chen, S. Gao, and Y. Ma, “Single-image crowd counting via multi-column convolutional neural network,” in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2016, pp. 589–597, doi: 10.1109/CVPR.2016.70.

[7] Y. Li, X. Zhang, and D. Chen, “CSRNet: Dilated convolutional neural networks for understanding the highly congested scenes,” in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2018, pp. 1091–1100, doi: 10.1109/CVPR.2018.00120.

[8] J. Ferryman and A. Shahrokni, “PETS2009: Dataset and challenge,” in Proc. 12th IEEE Int. Workshop Perform. Eval. Tracking Surveillance (PETS-Winter), 2009, pp. 1–6, doi: 10.1109/PETS-WINTER.2009.5399556.

[9] R. Hartley and A. Zisserman, Multiple View Geometry in Computer Vision, 2nd ed. Cambridge, U.K.: Cambridge University Press, 2004.

[10] H. Hotelling, “Stability in competition,” Econ. J., vol. 39, no. 153, pp. 41–57, 1929, doi: 10.2307/2224214.

[11] A. J. Heppenstall, A. T. Crooks, L. M. See, and M. Batty, Eds., Agent-Based Models of Geographical Systems. Dordrecht, The Netherlands: Springer, 2012, doi: 10.1007/978-90-481-8927-4.

[12] V. Grimm et al., “The ODD protocol: A review and first update,” Ecol. Model., vol. 221, no. 23, pp. 2760–2768, 2010, doi: 10.1016/j.ecolmodel.2010.08.019.

[13] P. J. Clark and F. C. Evans, “Distance to nearest neighbor as a measure of spatial relationships in populations,” Ecology, vol. 35, no. 4, pp. 445–453, 1954, doi: 10.2307/1931034.